



Software Requirements Specification (SRS)

Software Requirements Specification: A Contract Document

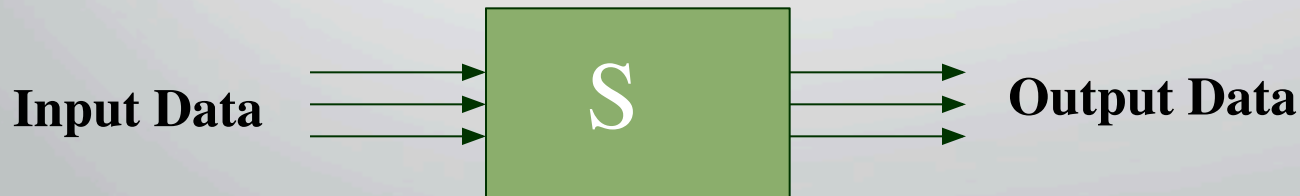
- Requirements document is a reference document.
- SRS document is a contract between the development team and the customer.
 - Once the SRS document is approved by the customer,
 - Any subsequent controversies are settled by referring the SRS document.

SW Requirements Specification

- Purpose of SRS
 - Communication between the Customer, Analyst, system developers, maintainers, ...
 - Contract between Purchaser and Supplier
 - Firm foundation for the design phase
 - Support system testing activities
 - Support project management and control
 - Controlling the evolution of the system

SRS Document (CONT.)

- The SRS document is also known as black-box specification:
 - the system is considered as a black box whose internal details are not known.
 - only its visible external (i.e. input/output) behaviour is documented.



SRS Document (CONT.)

- SRS document concentrates on:
 - What needs to be done
 - Carefully avoids the solution ("how to do") aspects.
- The SRS document serves as a contract
 - Between development team and the customer.
 - Should be carefully written

SRS Document (CONT.)

- The requirements at this stage:
 - Written using end-user terminology.
- If necessary:
 - Later a formal requirement specification may be developed from it.

What is **not** included in an SRS ?

- Project requirements
 - cost, delivery schedules, staffing, reporting procedures
- Design solutions
 - partitioning of SW into modules, choosing data structures
- Product assurance plans
 - Quality Assurance procedures, Configuration Management procedures, Verification & Validation procedures

Benefits of SRS

- Forces the users to consider their specific requirements carefully
- Enhances communication between the Purchaser and System developers
- Provides a firm foundation for the system design phase
- Enables planning of validation, verification, and acceptance procedures
- Enables project planning eg. estimates of cost and time, resource scheduling

Usable during maintenance phase

Types of Requirements

- Functional requirements
- Non functional requirements
 - Performance requirements
 - Interface requirements
 - Design constraints
 - Other requirements

SRS Standards

- ANSI/IEEE SRS Standard 830-1984
- BS 6719: 1986
- European Space Agency Standards
(ESA PSS-05-0, Jan 1987)
- US DoD-Std-7935A
- ...

Appendices

- Not always necessary
- It may include:
 - sample I/O formats
 - DFD, ERD documents
 - results of user surveys, cost analysis studies
 - supporting documents to help readers of SRS

Characteristics of a Good SRS

- Unambiguous
- Complete
- Verifiable
- Consistent
- Modifiable
- Traceable
- Usable during the Operation and Maintenance phase

SRS Review

- Formal Review done by Users, Developers, Managers, Operations personnel
- To verify that SRS confirms to the actual user requirements
- To detect defects early and correct them.
- Review typically done using checklists.